Worksheet 8.1

Science Quest 10: pages 308–10

Speed and velocity

Student: ..... Class: .....

Calculating average speed

1. The average speed (v) for a journey is calculated using the formula:

$$v = \frac{d}{t}$$

where d = distance travelled and t = time taken.

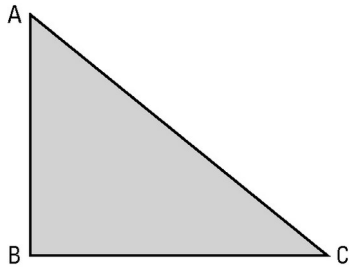
Complete the following table. Make sure you include the appropriate units for each quantity.

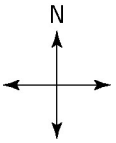
Distance travelled	Time taken	Average speed
100 m	20 s	
3 km		60 km/h
	15 s	20 m/s
2 km		25 m/s

Average speed and average velocity

2. Two vehicles (X and Y) travel from A to C along different paths as shown in the diagram (not drawn to scale). X travels along the path ABC and vehicle Y travels along AC. The triangle ABC is a right angle triangle. A is north of B.

AB = 600 m
BC = 800 m





Both vehicles take 70 s for the trip.
Perform the necessary calculations to complete the table.

Average speed of X	Average velocity of X	Average speed of Y	Average velocity of Y

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